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N + - Network Fundamentals & Building Networks.

Configuring OSPF.

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TASK 1.

❖ Using Cisco Packet Tracer, create a network topology with three routers in two OSPF areas (Area 0 and Area 1), and assign appropriate IP addresses to all interfaces.

- **Creating a Three-Router OSPF Network with Two Areas:**

Network Setup: Three routers were connected in Cisco Packet Tracer. Router1 and Router2 use Area 0, while Router2 and Router3 connect through Area 1.

IP Address Configuration:

Router Interface	IP Address	Subnet Mask	OSPF Area
R1 – S0/0/0	10.0.0.1	255.255.255.252	Area 0
R2 – S0/0/0	10.0.0.2	255.255.255.252	Area 0
R2 – S0/0/1	10.0.0.5	255.255.255.252	Area 1
R3 – S0/0/0	10.0.0.6	255.255.255.252	Area 1

Result: A three-router network was configured with two OSPF areas, with Router2 acting as the Area Border Router (ABR).

TASK 2.

❖ Configure OSPF on each router in your topology so that Area 0 acts as the backbone and Area 1 connects through one of the routers; verify OSPF neighbor relationships using the 'show ip ospf neighbor' command.

- **Configure OSPF in Area 0 and Area 1:**

Configure OSPF:
Router 1:

router ospf 1
network 10.0.0.0 0.0.0.3 area 0

Router 2:

router ospf 1
network 10.0.0.0 0.0.0.3 area 0
network 10.0.0.4 0.0.0.3 area 1

Router 3:

router ospf 1
network 10.0.0.4 0.0.0.3 area 1

Verify Neighbors:

Run this command on each router:

show ip ospf neighbor

Result: Area 0 acts as the OSPF backbone, while Area 1 connects through Router 2, which acts as the Area Border Router (ABR). The neighbor output confirms whether OSPF adjacencies are established.

TASK 3.

❖ Add two PCs to different areas in your OSPF network, assign them IP addresses, and test connectivity by pinging from one PC to the other.

- **Testing Connectivity Between PCs in Different OSPF Areas:**

IP Address Configuration:

Device	IP Address	Subnet Mask	Default Gateway
PC1 (Area 0)	192.168.1.10	255.255.255.0	192.168.1.1
PC2 (Area 1)	192.168.2.10	255.255.255.0	192.168.2.1

Connectivity Test:
From PC1, run:

ping 192.168.2.10

Result: Successful ping replies confirm that both PCs can communicate through the OSPF network.

TASK 4.

❖ Simulate a router failure by shutting down the interface connecting Area 1 to Area 0, then use 'show ip route' and 'show ip ospf' commands to observe how OSPF reconverges and restores connectivity after re-enabling the interface, Track the time it takes for OSPF to update the routing tables after the failure and recovery.

- **OSPF Failure, Recovery and Convergence:**

Simulate Failure: The interface connecting Area 1 to Area 0 was shut down**:**

Router(config)# interface serial0/0/1
Router(config-if)# shutdown

Observe Routing Tables: The following commands were used to check OSPF and routing updates.

show ip route
show ip ospf

Recovery: The interface was re enabled using the no shutdown command. OSPF re established its neighbor relationship and updated the routing table.

Observation: The unreachable routes were removed after failure, and connectivity was restored after recovery. The convergence time depends on OSPF timers and network conditions; the actual time should be recorded